

FYKE SIMON V. TONEL

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Education

STI College Caloocan

Caloocan City, Metro Manila

Bachelor of Science in Computer Science. GWA: 1.27 ($\approx 3.9/4.0$ equivalent)

Expected 2026

- President's Lister 2022-Present, Alumni Scholar (50% merit)
- Best Thesis & Best Research Project, ICTE Symposium (campus-wide CS)

Experience

Back-End Developer (Intern) – Inspire Holdings Inc.

January 2026 – February 2026

- Built and deployed a NestJS payment backend on AWS (Lambda, API Gateway, RDS), integrating secure UnionBank APIs.
- Reduced **p95** latency from **420ms to 180ms (-57%)** and CPU usage by **35%** using Redis caching and token-bucket rate limiting
- Enforced **API security via HMAC signing** and request validation.
- Delivered **CI/CD pipelines** and **40+ unit tests** using GitHub Actions and Jest.

Software Developer (Intern) – Socia I.T. Solutions

February 2026 – March 2026

- Migrated a web-based streaming platform to React Native (Expo), consolidating iOS and Android into a unified codebase.
- Implemented WebSocket-driven real-time features for live streaming and interactive rewards.
- Strengthened runtime reliability by enforcing TypeScript strict mode and structured validation.
- Contributed to scalable frontend architecture for low-latency mobile applications.

Projects

Flick Player: Music Player with Custom UAC 2.0

January 2026

Flutter, Dart, Rust, Kotlin, DSP

- Engineered a **Rust-based audio engine** enabling direct USB output and bypassing Android audio pipeline constraints.
- Designed a **four-engine architecture** to ensure compatibility across devices and playback modes.
- Implemented **real-time audio processing pipeline** with low-latency performance.
- Open-source project with **600+ installs/downloads**.

br41ndmg: High-Fidelity Audio Resampling Library

January 2026

Rust (DSP, Audio Processing)

- Implemented polyphase sinc-based resampling for high-precision sample rate conversion.
- Designed configurable **FIR filters** achieving **>100 dB** stopband attenuation and **<0.1 dB passband ripple**.

PASADA: Ride-Hailing & Fleet Management Ecosystem

December 2024 – November 2025

Flutter, TypeScript (Bun/Deno), React, Supabase, Google Cloud Platform

- Architected a distributed system spanning **4 applications** (Passenger, Driver, Admin, Web) with shared authentication and RBAC.
- Built a real-time tracking system using MapLibre GL and OpenStreetMap for live fleet monitoring.
- Developed a data pipeline for **7-day demand forecasting** and analytics.
- Integrated AI-powered summaries to surface actionable insights from real-time transport data.

Technical Skills

Languages: Rust, TypeScript, Dart, Kotlin, Java, SQL

Backend: Node.js, NestJS, PostgreSQL, Redis, Supabase

Frontend/Mobile: Flutter, React Native, React, Next.js

Cloud/DevOps: AWS, GCP, Docker, Cloudflare, CI/CD

Systems: WebSockets, API design, rate limiting, real-time systems